Department of Economics
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Problem Set 1
Macroeconomics I
Due before class, Monday, November 2, 2026

Recall the single-sector neoclassical model with search from class. In this problem set, we solve the planner version of this model using a linear approximation to the model around its non-stochastic steady state.

In later problem sets, we use a shooting algorithm to solve the model equations exactly under the assumption of perfect foresight, and then use a value function method and a projection method to solve the model globally under uncertainty.

The parameters you should use in this problem set are listed below. We'll call these the "baseline parameters" going forward. Rather than setting the matching function level χ directly, we target the steady-state unemployment rate $u = 1 - N^{ss}$ and back out the value of χ consistent with that target.

Table 1: Numerical Parameter Values

Concept	Symbol	Value
Discount factor	β	0.99
Inverse IES	σ	2.00
Capital share	α	0.30
Capital depreciation	δ_k	0.03
Labor separation	δ_n	0.10
Vacancy cost	ϕ_n	0.50
Steady-state unemployment	u	0.06
Matching function elasticity	ε	0.25
$\log A$ persistence	ρ	0.95
$\log A$ shock s.d.	σ_a	0.01

Note: in building your code, you are free to use and modify the example programs on the course website. *See the syllabus for the course policy on AI tools.*

Please provide your solution as a single .zip file. The file should contain all of the code you used for your solution, run by a base call `main_prog`, and a typeset responses document with all specifically requested answers. Call this file `responses_<lastname>.pdf`. Your grade will be primarily based on this .pdf document of your responses. Requested answers are highlighted by **boldface** words or sentences in most questions. Please include figures inline in your document, as you might see in an academic paper. When only numbers are requested, you may report those numbers without comment.

1. **Solution to linearized model:**

   (a) Using the model equations from our planner economy, create a function called `model_ss` that takes as an input a data structure called `param` of parameter values, and returns as an output the implied value of χ and a vector containing the steady-state values of the variables $[X^{ss}, Y^{ss}] \equiv [[A, K, N], [Y, C, I, N, V]]$. Using the parameters in Table 1, your program should return the values in Table 2, including the implied value of χ.

Table 2: Steady-state values and implied parameter

Variable	Value
A	1.0000
K	16.6591
N	0.9400
Y	2.2268
C	X.XXXX
I	0.4998
N	0.9400
V	0.1508
Implied parameter	
χ	0.6136

   To confirm your solution for the steady-state, please **report** the missing value for steady-state consumption. Also, **explain** why the variable N appears in both the jump and state vectors in this table.

   (b) Create a second function, called `model`, which takes as input the data structure called `param` and returns the first-order derivatives of $F(x, y)$ evaluated at the steady-state F_x, F_y, F_{xp}, F_{yp}, where $y = \log(Y)$, $x = \log(X)$, etc. The function `model` should call the function `model_ss` you wrote above.

   To confirm your solution, please **report** the derivative of the resource constraint equation with respect to $\log(C_t)$.

(c) Using the code `gx_hx`, generate the model "solution matrices" g_x, h_x, and η. Below, I provide my solution for h_x; please **report** corresponding values of g_x and η.

$$h_x = \begin{bmatrix} 0.9500 & 0 & 0 \\ 0.0903 & 0.9677 & 0.0545 \\ 0.0250 & 0.0084 & 0.3487 \end{bmatrix}$$

(d) Using your solution to 1(c) above, compute impulse responses to a one standard deviation disturbance to technology for each of the eight variables listed above.

Generate and provide a 4x2 figure in Matlab. In each subplot, plot the impulse response of one of the eight main model variables according to the approximate solution of the model. Your plot should include the impact through 20 periods after the shock. Plot the variables in the order they appear in part 1(a) of the problem set, and label each subplot in the figure with the variable name.

(e) Create a local random number stream with `rs = RandStream('twister','Seed',0)` and use it to draw 5000 standard normal scalar shocks $\{\epsilon_t\}$ with `randn(rs,5000,1)`. Using your solution from 1(c) above and the generated shocks, simulate 5001 periods of data from the model. Initialize all state variables in the first period of your simulation to be zero. **Report** the first 5 realizations of log productivity in your simulations, and **reproduce** Table 3, adding the missing values.

Table 3: Moments from simulated linear model

Moment	Model	Data
Std $\log(Y)$	0.0452	0.0149
Std $\log(C)$		0.0082
Std $\log(I)$	0.0964	0.0318
Std $\log(N)$		0.0131
Autocorr $\log(Y)$		0.85
Autocorr $\log(C)$		0.85
Autocorr $\log(I)$		0.84
Autocorr $\log(N)$		0.93
Std $\log(N)$ / Std $\log(Y)$		0.88

(f) One common criticism of labor search models is that they cannot match the relative volatility of output and employment in the data. **Report** the ratio of the two standard deviations $std(N)/std(Y)$ in the model and the data. Find and **report** a parameterization that makes this ratio substantially closer to the data. **Describe** the key features of this parameterization, and attempt to **provide intuition** for why this works.